

3.5 Shapes of molecules homework — answers

CIE 9701 AS Chemistry · 8 multiple-choice + 10 written marks

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Section A — Multiple choice

Quick answers

Q	Ans.	Q	Ans.	Q	Ans.
A1	A	A4	B	A7	B
A2	D	A5	B	A8	D
A3	C	A6	C		

Worked explanations

A1. A

2025 MJ Paper 12 Q7

The bond angles to use are H_2O 104.5° , NH_3 107° and BF_3 120° . So the *smallest* angle belongs to H_2O (two lone pairs on O squeeze the H–O–H angle most) and the *largest* to BF_3 (three bond pairs, no lone pair, 120°). Row A gives H_2O as smallest and BF_3 as largest.

B / C / D: each puts the species in the wrong order — e.g. H_2O is never the largest angle, because its two lone pairs reduce the angle below the tetrahedral value.

A2. D

2023 MJ Paper 13 Q5

$\text{NH}_3 + \text{H}^+ \longrightarrow \text{NH}_4^+$. In NH_3 , nitrogen has 3 bond pairs *and* 1 lone pair (pyramidal, angle about 107°). When the lone pair is used to bond to H^+ , the nitrogen in NH_4^+ has **four bond pairs and no lone pair**. Four identical bond pairs give a *tetrahedral* shape with bond angle **109.5°** . So the row “tetrahedral, 109.5° ” is correct.

A / B: say the shape is pyramidal. NH_4^+ has no lone pair, so it is not pyramidal.

C: gives the right shape but keeps the NH_3 angle of 107° ; removing the lone pair opens the angle to 109.5° .

A3. C

2024 MJ Paper 11 Q24

The ammonium ion, NH_4^+ , has four bonding pairs of electrons around nitrogen and no lone pair, so it is tetrahedral with a bond angle of **109.5°** .

A: 90° is an octahedral angle, not tetrahedral.

B: 107° is the angle in NH_3 (one lone pair), not in NH_4^+ .

D: 120° is a trigonal planar angle (e.g. BF_3).

A4. B

2025 MJ Paper 14 Q11

PF_5 has five bonding pairs around phosphorus and *no lone pair*, so it is trigonal bipyramidal. This makes statement B correct.

A: not all F–P–F angles are 90° . The two axial F atoms make 90° with the equatorial F atoms, but the equatorial F–P–F angles are 120° .

C: PF_5 is symmetrical, so the bond dipoles cancel and the molecule has *no* overall dipole moment.

D: the shape is trigonal bipyramidal, not octahedral (octahedral is 6 bond pairs, e.g. SF_6).

A5. B

2021 MJ Paper 11 Q34

(1) C_2H_4 (ethene) — the $\text{H}-\text{C}-\text{H}$ angle is about 117° and the $\text{H}-\text{C}=\text{C}$ angle about 121° , so C_2H_4 has angles close to 120° . **Correct.** (2) PF_5 — trigonal bipyramidal, so the equatorial $\text{F}-\text{P}-\text{F}$ angles are exactly 120° . **Correct.** (3) NCl_3 — nitrogen has 3 bond pairs and 1 lone pair, giving a pyramidal shape with angles of about 107° ; it has no 120° angle. **Wrong.** So statements 1 and 2 only are correct.

A: includes NCl_3 , which has no 120° angle.

C: drops C_2H_4 , which does have a 120° angle.

D: keeps only statement 1 and loses the correct statement about PF_5 .

A6. C

2025 MJ Paper 14 Q3

Compare the smallest bond angle in each pair: A CH_4 109.5° vs SF_6 90° (smallest) — 109.5 is *not* smaller than 90 . B CO_2 180° vs BF_3 120° — 180 is not smaller than 120 . C H_2O 104.5° vs H_3O^+ 107° — 104.5 is smaller than 107 . **Correct.** D NH_4^+ 109.5° vs NH_3 107° — 109.5 is not smaller than 107 .

A / B / D: in each case the first species has the larger (or equal) bond angle, so the required “smaller in the first species” condition fails.

A7. B

2022 MJ Paper 13 Q6

In hydrazine, $\text{H}_2\text{N}-\text{NH}_2$, each nitrogen atom has **3 bond pairs and 1 lone pair** (2 $\text{N}-\text{H}$ bonds, 1 $\text{N}-\text{N}$ bond, 1 lone pair). This is the same electron-pair arrangement as in NH_3 , so the angle X is the pyramidal angle of about **107°** .

A: 90° would require octahedral electron-pair geometry.

C: 109.5° is the tetrahedral angle, which needs *four* bond pairs and no lone pair — not the case at nitrogen here.

D: 120° is trigonal planar (three bond pairs, no lone pair).

A8. D

2024 MJ Paper 13 Q6

Borazole, $\text{N}_3\text{B}_3\text{H}_6$, is a six-membered ring of alternating N and B atoms — the “inorganic benzene”. Like benzene it is **planar**, and its delocalised π system contains **6 π electrons** (the aromatic 6- π -electron count).

A / B: say borazole is non-planar; it is planar, like benzene.

C: gives 3 π electrons; borazole is aromatic with 6 π electrons.

Section B — Mark scheme clips

B1(a)(iii) [1]

1(a)(iii)	180°	1
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B2(b)(i) [2]

2(b)(i)	M1 1 shared pair of electrons between each S-Cl shown as •• M2 6 non-bonding electrons around each Cl shown as × AND 4 non-bonding electrons around S shown as •	2
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B2(b)(ii) [2]

2(b)(ii)	M1 shape non-linear M2 bond angle 103–105(°)	2
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B4(e) [2]

1(e)	M1 6 bonding pairs (of electrons and 0 lone pairs) OR coordination number = 6 OR x = 6 M2 bonding pairs repel (equally)	2
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B5(b)(ii) [1]

1(b)(ii)	trigonal bipyramid(al)	1
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B6(b)(i) [2]

4(b)(i)	M1 non-linear M2 bond angle – 120 > × ≥ 116°	2
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